



Mark Scheme (Results)

October 2021

Pearson Edexcel International Advanced
Subsidiary Level in Physics (WPH11)
Paper 01 Mechanics and Materials

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

1. Mark scheme format

- 1.1 You will not see 'wtte' (words to that effect). Alternative correct wording should be credited in every answer unless the MS has specified specific words that must be present. Such words will be indicated by underlining e.g. 'resonance'
- 1.2 Bold lower case will be used for emphasis e.g. '**and**' when two pieces of information are needed for 1 mark.
- 1.3 Round brackets () indicate words that are not essential e.g. "(hence) distance is increased".
- 1.4 Square brackets [] indicate advice to examiners or examples e.g. [Do not accept gravity] [ecf].

2. Unit error penalties

- 2.1 A separate mark is not usually given for a unit but a missing or incorrect unit will normally mean that the final calculation mark will not be awarded.
- 2.2 This does not apply in 'show that' questions or in any other question where the units to be used have been given, for example in a spreadsheet.
- 2.3 The mark will not be awarded for the same missing or incorrect unit only once within one clip in open.
- 2.4 Occasionally, it may be decided not to insist on a unit e.g. the candidate may be calculating the gradient of a graph, resulting in a unit that is not one that should be known and is complex.
- 2.5 The mark scheme will indicate if no unit error is to be applied by means of [no ue].

3. Significant figures

- 3.1 Use of too many significant figures in the theory questions will not prevent a mark being awarded if the answer given rounds to the answer in the MS.
- 3.2 Too few significant figures will mean that the final mark cannot be awarded in 'show that' questions where one more significant figure than the value in the question is needed for the candidate to demonstrate the validity of the given answer.
- 3.3 The use of one significant figure might be inappropriate in the context of the question e.g. reading a value off a graph. If this is the case, there will be a clear indication in the MS.
- 3.4 The use of $g = 10 \text{ m s}^{-2}$ or 10 N kg^{-1} instead of 9.81 m s^{-2} or 9.81 N kg^{-1} will mean that one mark will not be awarded. (but not more than once per clip). Accept 9.8 m s^{-2} or 9.8 N kg^{-1}
- 3.5 In questions assessing practical skills, a specific number of significant figures will be required e.g. determining a constant from the gradient of a graph or in uncertainty calculations. The MS will clearly identify the number of significant figures required.

4. Calculations

- 4.1 Bald (i.e. no working shown) correct answers score full marks unless in a 'show that' question.
- 4.2 If a 'show that' question is worth 2 marks. then both marks will be available for a reverse working; if it is worth 3 marks then only 2 will be available.
- 4.3 **use** of the formula means that the candidate demonstrates substitution of physically correct values, although there may be conversion errors e.g. power of 10 error.
- 4.4 **recall** of the correct formula will be awarded when the formula is seen or implied by substitution.
- 4.5 The mark scheme will show a correctly worked answer for illustration only.

5. Quality of Written Communication

- 5.1 Indicated by QoWC in mark scheme. QWC – Work must be clear and organised in a logical manner using technical wording where appropriate.
- 5.2 Usually it is part of a max mark, the final mark not being awarded unless the QoWC condition has been satisfied.

6. Graphs

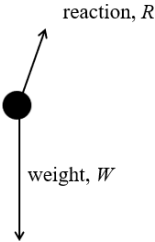
- 6.1 A mark given for axes requires both axes to be labelled with quantities and units, and drawn the correct way round.
- 6.2 Sometimes a separate mark will be given for units or for each axis if the units are complex. This will be indicated on the mark scheme.
- 6.3 A mark given for choosing a scale requires that the chosen scale allows all points to be plotted, spreads plotted points over more than half of each axis and is not an awkward scale e.g. multiples of 3, 7 etc.
- 6.4 Points should be plotted to within 1 mm.
 - Check the two points furthest from the best line. If both OK award mark.
 - If either is 2 mm out do not award mark.
 - If both are 1 mm out do not award mark.
 - If either is 1 mm out then check another two and award mark if both of these OK, otherwise no mark.For a line mark there must be a thin continuous line which is the best-fit line for the candidate's results.

Question Number	Answer	Mark
1	B is the correct answer A is incorrect because gradient is velocity, not acceleration C is incorrect because gradient is rate of change of acceleration, not velocity D is incorrect because gradient is rate of change of acceleration, not displacement	1
2	C is the correct answer A is incorrect because magnitudes are equal B is incorrect because magnitudes are equal and directions are opposite D is incorrect because directions are opposite	1
3	C is the correct answer A is incorrect because reducing the extension reduces the stored energy B is incorrect because reducing the extension reduces the stored energy and increasing the force increases the stored energy D is incorrect because increasing the force increases the stored energy	1
4	C is the correct answer A is incorrect because $13.2/\text{answer} = \cos 22.6^\circ$, not $\sin 22.6^\circ$ B is incorrect because $13.2/\text{answer} = \cos 22.6^\circ$, not $\sin 22.6^\circ$, and the equation has also been incorrectly transposed. D is incorrect because although $5.5/\text{answer} = \sin 22.6^\circ$ the equation has been incorrectly transposed.	1
5	B is the correct answer A is incorrect because the CW moment of the force could balance the ACW moment of the weight C is incorrect because the CW moment of the force could balance the ACW moment of the weight D is incorrect because the CW moment of the force could balance the ACW moment of the weight	1

6	A is the correct answer B is incorrect because N is in the wrong direction C is incorrect because the wrong diagonal has been used D is incorrect because the wrong diagonal has been used	1
7	A is the correct answer B is incorrect because the distance has not been doubled and the time has not been squared C is incorrect because the distance is in the wrong unit D is incorrect because the distance is in the wrong unit, has not been doubled and the time has not been squared	1
8	B is the correct answer A is incorrect because the factor $9.81 = g / (\text{N kg}^{-1})$ is not needed since it is already included in the weight, $\Delta E_{\text{grav}} = mg\Delta h = W\Delta h$ C is incorrect because dividing by 9.81 removes g from the calculation D is incorrect because dividing by 9.81 removes g and the factor $25 = \Delta h / \text{m}$ is missing from the calculation	1
9	A is the correct answer B is incorrect because low viscosity is not a condition C is incorrect because cylindrical object is not a condition D is incorrect because neither cylindrical object nor low viscosity are conditions	1
10	D is the correct answer A is incorrect because mg has the wrong direction B is incorrect because mg and ma have the wrong directions C is incorrect because ma has the wrong direction	1
Total for Section A		10

Question Number	Answer	Mark
11(a)	Use of $E_k = \frac{1}{2} m v^2$ (1) $E_k = 0.54 \text{ J}$ (1) <u>Example of calculation</u> $E_k = 0.5 \times 0.16 \text{ kg} \times (2.6 \text{ m s}^{-1})^2 = 0.541 \text{ J}$	2
11(b)	Use of $E_{\text{grav}} = m g \Delta h$ (1) $\Delta h = 0.51 \text{ m}$ (1) (allow ecf from (a)) <u>Example of calculation</u> Decrease in GPE = $0.54 \text{ J} + 0.26 \text{ J} = 0.8 \text{ J}$ $\Delta h = 0.8 \text{ J} / (0.16 \text{ kg} \times 9.81 \text{ m s}^{-2}) = 0.51 \text{ m}$	2
	Total for question 11	4

Question Number	Answer	Mark
12(a)	the total momentum before (a collision) = the total momentum after (a collision) Or Sum of momentum values before (collision) = sum of momentum values after (collision) Or total momentum remains constant Or (1) the momentum of a system remains constant Provided no external/unbalanced/resultant force acts (on the system) Or (1) in a closed/isolated system	2
12(b)	Use of $p = m v$ (1) Uses conservation of momentum (1) Velocity = -4.6 m s^{-1} (1) <u>Example of calculation</u> $2.7 \text{ kg} \times 10 \text{ m s}^{-1} = 2.7 \text{ kg} \times v + 7.9 \text{ kg} \times 5.0 \text{ m s}^{-1}$ $v = (27.0 - 39.5) \text{ kg m s}^{-1} \div 2.7 \text{ kg} = -4.6(3) \text{ m s}^{-1}$	3
	Total for question 12	5

Question Number	Answer	Mark
13(a)	Vertical downwards force labelled "weight", or W. (1) Force perpendicular to slope labelled "reaction", "(normal) contact", or R or N. (1) 	2
13(b)(i)	Resolves acceleration along slope. (1) Acceleration = 1.2 m s^{-2} (1) <u>Example of calculation</u> $a = 9.81 \text{ m s}^{-2} \times \sin 6.9^\circ = 1.18 \text{ m s}^{-2}$	2
13(b)(ii)	Either Use of $v^2 = u^2 + 2 a s$ (1) Final speed = 12 m s^{-1} (ecf from (i)) (1) Or Use of $E_k = \frac{1}{2}mv^2$ and $\Delta E_{\text{grav}} = mg\Delta h$ (1) Final speed = 12 m s^{-1} (ecf from (i)) (1) <u>Example of calculation</u> $v^2 = 0^2 + 2 \times 1.18 \text{ m s}^{-2} \times 60 \text{ m}$ $v = \sqrt{(0 + 2 a s)} = \sqrt{(2 \times 1.18 \text{ m s}^{-2} \times 60 \text{ m})} = 11.9 \text{ m s}^{-1}$	2
13(b)(iii)	Use of $v = u + a t$ or another valid <i>suvat</i> equation (1) Time = 10 s (ecf from (ii)) (ecf from (i)) (1) <u>Example of calculation</u> $v = u + a t, u = 0$ $11.9 = 0 + 1.18 \text{ m s}^{-2} \times t$ $t = 11.9 \text{ m s}^{-1} \div 1.18 \text{ m s}^{-2} = 10.1 \text{ s}$	2
Total for question 13		8

Question Number	Answer	Mark
14(a)	Maximum value of weight/force for which weight/force is proportional to extension Or Point beyond which Hooke's Law no longer applies Or Point beyond which graph line ceases to be straight Or Point beyond which weight/force is no longer proportional to extension (1)	1
14(b)(i)	Use of large triangle to determine gradient (1) Gradient = 18 500 (N m⁻¹) (sf range 18 - 19, no ue) (1) <u>Example of calculation</u> gradient = 37 N ÷ (2 × 10⁻³ m) = 18 500 (N m⁻¹)	2
14(b)(ii)	Rearranges $E = \text{stress} / \text{strain}$ to get $E = \text{gradient} \times \frac{x}{A}$ Or Rearranges $E = \text{stress} / \text{strain}$ to get gradient = $\frac{A}{x} E$ (1) Use of $A = \pi r^2$ (1) Young modulus = 2 × 10¹¹ Pa (1) (allow ecf from (b)(i)) <u>Example of calculation</u> $A = \pi \times (2.8 \times 10^{-4})^2 = 2.46 \times 10^{-7} \text{ m}^2$ $E = 1.85 \times 10^4 \text{ N m}^{-1} \times 2.6 \text{ m} \div 2.46 \times 10^{-7} \text{ m}^2 = 1.95 \times 10^{11} \text{ Pa}$	3
14(c)	Use of $\sigma = \frac{F}{A}$ (1) Determines maximum safe load Or Determines maximum stress Or Determines minimum cross section (1) Valid conclusion by comparison with student's calculation (1) <u>Example of calculation</u> $\sigma_{\text{max}} = \frac{W_{\text{max}}}{A}$ $4.80 \times 10^8 \text{ Pa} = \frac{W_{\text{max}}}{2.46 \times 10^{-7} \text{ m}^2}$ $W_{\text{max}} = 480 \times 10^6 \text{ Pa} \times 2.46 \times 10^{-7} \text{ m}^2 = 118 \text{ N} > 100 \text{ N}$ so yes	3
	Total for question 14	9

Question Number	Answer	Mark																																						
15(a)*	This question assesses a student's ability to show a coherent and logically structured answer with linkages and fully-sustained reasoning. Marks are awarded for indicative content and for how the answer is structured and shows lines of reasoning. The following table shows how the marks should be awarded for indicative content and lines of reasoning. <table><tr><th>IC points</th><th>IC mark</th><th>Max linkage mark available</th><th>Max final mark</th></tr><tr><td>6</td><td>4</td><td>2</td><td>6</td></tr><tr><td>5</td><td>3</td><td>2</td><td>5</td></tr><tr><td>4</td><td>3</td><td>1</td><td>4</td></tr><tr><td>3</td><td>2</td><td>1</td><td>3</td></tr><tr><td>2</td><td>2</td><td>0</td><td>2</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table> <table><tr><td></td><td>Marks</td></tr><tr><td>Answer shows a coherent and logical structure with linkages and fully sustained lines of reasoning demonstrated throughout.</td><td>2</td></tr><tr><td>Answer is partially structured with some linkages and lines of reasoning</td><td>1</td></tr></table> Indicative content: 1. There is a (backward) force/friction on floor from wheels/car 2. Newton's Third Law implies forward/opposite force from floor 3. Compression/deformation of spring reduces (as car moves forward) 4. (Resultant) force is proportional to compression/deformation of spring Or Reference to Hooke's law. 5. Acceleration is proportional to resultant force Or Reference to $F = ma$ 6. Acceleration reduces (as distance travelled increases) Or Acceleration is zero once spring has returned to original state Accept "resultant force" for "acceleration" in IC6	IC points	IC mark	Max linkage mark available	Max final mark	6	4	2	6	5	3	2	5	4	3	1	4	3	2	1	3	2	2	0	2	1	1	0	1	0	0	0	0		Marks	Answer shows a coherent and logical structure with linkages and fully sustained lines of reasoning demonstrated throughout.	2	Answer is partially structured with some linkages and lines of reasoning	1	6
IC points	IC mark	Max linkage mark available	Max final mark																																					
6	4	2	6																																					
5	3	2	5																																					
4	3	1	4																																					
3	2	1	3																																					
2	2	0	2																																					
1	1	0	1																																					
0	0	0	0																																					
	Marks																																							
Answer shows a coherent and logical structure with linkages and fully sustained lines of reasoning demonstrated throughout.	2																																							
Answer is partially structured with some linkages and lines of reasoning	1																																							
Total for question 15		6																																						

Question Number	Answer	Mark
16(a)	Resolves velocity to find vertical component (1) Use of $v = u + at$ (1) Time to max height = 3.3 (s) (1) <u>Example of calculation</u> $u_v = 50 \text{ m s}^{-1} \sin(40^\circ) = 32.1 \text{ m s}^{-1}$ $v = u + at$ with $v = 0$ $t = 32.1 \text{ m s}^{-1} \div 9.81 \text{ m s}^{-2} = 3.28 \text{ s}$	3
16(b)	Use of $v_H = v \cos \theta$ (1) Use of $s = ut$ to calculate horizontal distance (1) Use of $s = ut + \frac{1}{2}at^2$ to calculate maximum height or other <i>suvat</i> equation (1) Maximum height = 53 m (allow ecf from (a)) (1) Use of $\tan \theta$ to calculate vertical height of hill (1) Compares height of hill with maximum height (accept conclusion based on candidate's values). Or Use of $v_H = v \cos \theta$ (1) Use of $s = ut$ to calculate horizontal distance (1) Use of $s = ut + \frac{1}{2}at^2$ to calculate maximum height or other <i>suvat</i> equation (1) Maximum height = 53 m (allow ecf from (a)) (1) Use of $\tan \theta$ to calculate minimum angle of hill for a hit (1) Compares angle of hill with minimum angle (accept conclusion based on candidate's values). Or (1) Use of $v_H = v \cos \theta$ (1) Use of $s = ut$ to calculate horizontal distance (1) Use of $s = ut + \frac{1}{2}at^2$ to calculate maximum height or other <i>suvat</i> equation (1) Maximum height = 53 m (allow ecf from (a)) (1) Use of $\tan \theta$ to calculate horizontal distance to 52.6 m height along hill Compares horizontal distances (accept conclusion based on candidate's values). <u>Example of calculation</u> $v_H = 50 \text{ m s}^{-1} \times \cos(40^\circ) = 38.3 \text{ m s}^{-1}$ $s_H = v_H \times t$ $s_H = 38.3 \text{ m s}^{-1} \times 3.28 \text{ s} = 125 \text{ m}$ maximum height of rock = $s_v = u_v t + \frac{1}{2}at^2$ with $a = -g$	6

	$s_v = 32.1 \text{ m s}^{-1} \times 3.28 \text{ s} - \frac{1}{2} \times 9.81 \text{ m s}^{-2} \times (3.28 \text{ s})^2 = 52.6 \text{ m}$ vertical height of hill at horizontal distance of 125 m = $125 \text{ m} \times \tan(20^\circ)$ = 45.7 m 45.7 m < 52.6 m, so no	
	Total for question 16	9

Question Number	Answer	Mark
17(a)(i)	The layers of fluid flow past each other without mixing Or <u>Velocity</u> at a fixed point (relative to the drop) remains constant (1)	1
17(a)(ii)	Resultant force is zero Or Sum of the vertical forces is zero (1) (Accept $W - U = D$ or $W = U + D$ with terms defined)	1
17(b)(i)	Use of $\rho = \frac{m}{V}$ (1) Use of $W = mg$ (1) Weight = 3.3×10^{-4} N (1) <u>Example of calculation</u> $1.00 \times 10^3 \text{ kg m}^{-3} = m \div 3.35 \times 10^{-8} \text{ m}^3$ $m = 1.00 \times 10^3 \text{ kg m}^{-3} \times 3.35 \times 10^{-8} \text{ m}^3 = 3.35 \times 10^{-5} \text{ kg}$ $W = mg = 3.35 \times 10^{-5} \text{ m}^3 \times 9.81 \text{ N kg}^{-1} = 3.29 \times 10^{-4} \text{ N}$	3
17(b)(ii)	Use of upthrust = weight of fluid displaced (1) Upthrust = 3.1×10^{-4} (N) (1) <u>Example of calculation</u> $0.94 \times 10^3 \text{ kg m}^{-3} = m \div 3.35 \times 10^{-8} \text{ m}^3$ $m = 0.94 \times 10^3 \text{ kg m}^{-3} \times 3.35 \times 10^{-8} \text{ m}^3 = 3.15 \times 10^{-5} \text{ kg}$ $U = mg = 3.15 \times 10^{-5} \text{ m}^3 \times 9.81 \text{ N kg}^{-1} = 3.09 \times 10^{-4} \text{ N}$	2
17(b)(iii)	Uses upthrust and weight to determine the viscous force F (1) Use of $V = \frac{4}{3} \pi r^3$ to determine r (1) Use of $F = 6\pi\eta rv$ (1) $v = 4.8 \times 10^{-3} \text{ m s}^{-1}$ (ecf from (b)(i) and (b)(ii)) (1) <u>Example of calculation</u> $W = U + 6\pi\eta rv \rightarrow W - U = 6\pi\eta rv$ $W - U = (3.29 - 3.09) \times 10^{-4} \text{ N} = 2.0 \times 10^{-5} \text{ N}$ $r = \sqrt[3]{\frac{3}{4} \times 3.35 \times 10^{-8} \text{ m}^3 \div \pi} = 2.0 \times 10^{-3} \text{ m}$ $2.0 \times 10^{-5} \text{ N} = 6\pi \times 0.11 \text{ Pa s} \times 2.0 \times 10^{-3} \text{ m} \times v$ $v = 2.0 \times 10^{-5} \text{ N} \div (6\pi \times 0.11 \text{ Pa s} \times 2.0 \times 10^{-3} \text{ m}) = 4.82 \times 10^{-3} \text{ m s}^{-1}$	4
	Total for question 17	11

Question Number	Answer	Mark
18(a)	Use of $P = W / t$ and $\Delta W = F \Delta s$ (1) Force = 13.9 (kN) (1) <u>Example of calculation</u> In 1 second $W = 6250$ J and distance travelled = 0.450 m $F = 6250 \text{ W} \div 0.450 \text{ m s}^{-1} = 13.9 \text{ kN}$	2
18(b)	Use of $\Delta W = F \Delta s$ (1) Use of $\Delta s = 4.35 / \sin 6.0^\circ$ (1) Total work = 5.8×10^5 J (allow ecf from (a)) (1) <u>Example of calculation</u> $\Delta W = 13.9 \times 10^3 \text{ N} \times 4.35 \text{ m} \div \sin 6.0^\circ = 578 \text{ kJ}$	3
18(c)	Use of $\Delta E_{\text{grav}} = m g \Delta h$ (1) Useful work done = 89.6 (kJ) (1) <u>Example of calculation</u> $\Delta E_{\text{grav}} = 2.10 \times 10^3 \text{ kg} \times 9.81 \text{ N kg}^{-1} \times 4.35 \text{ m} = 89.6 \text{ kJ}$	2
18(d)	Use of $\varepsilon = \text{useful energy output} / \text{total energy input}$ (1) Efficiency = 0.16 (allow ecf from (b) and (c)) (1) <u>Example of calculation</u> $\varepsilon = 89.6 \text{ kJ} \div 578 \text{ kJ} = 0.155$	2
	Total for question 18	9

Question Number	Answer	Mark
19(a)	Use of moment = $F \times$ (1) Anticlockwise moment = 25.2 (Nm) and (maximum) clockwise moment = 24.2 (Nm) (1) As angle to the ground increases, clockwise moment from the weight decreases Or (1) If line of action of weight moves outside base cannot regain equilibrium. (1) $25.2 > 24.2 \therefore$ blows over <u>Example of calculation</u> moment from wind = $14 \text{ N} \times 1.8 \text{ m} = 25.2 \text{ N m}$ moment from weight = $110 \text{ N} \times 0.22 \text{ m} = 24.2 \text{ N m}$ $25.2 > 24.2 \therefore$ blows over	4
19(b)	Horizontal component = $T \times \sin 44^\circ$ Or Distance to line of action of $T = 1.5 \times \sin 44^\circ$ (1) Equates clockwise to anticlockwise moments about centre of base to determine T (1) Use of trigonometry to calculate vertical component of tension (1) Adds weight to vertical component Force exerted on the ground = 141 N (1) (1) <u>Example of calculation</u> Horizontal component of tension = $T \times \sin 44^\circ$ CWM = $1.5 \text{ m} \times T \times \sin 44^\circ = 1.04 \text{ m} \times T$ ACWM = $25 \text{ N} \times 1.8 \text{ m} = 45.0 \text{ N m}$ $1.04 \text{ m} \times T = 45.0 \text{ N m}$ $T = 45.0 \text{ N m} \div 1.04 \text{ m} = 43.2 \text{ N}$ Vertical component of $T = 43.2 \text{ N} \times \cos 44^\circ = 31.1 \text{ N}$ Total downward force = $110 \text{ N} + 31.1 \text{ N} = 141.1 \text{ N}$	5
	Total for question 19	9